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Aryabhatta Maths Olympiad (AMO)

Level 1 — Practice Paper 4

CLASS 7

Questions	45 multiple-choice questions
Marks	100
Time	60 minutes
Section I	40 questions × 2 marks = 80 marks
Section II	5 Star Questions × 4 marks = 20 marks
Negative marking	No negative marking
Calculator	Not allowed
Chapters	Number System · Operations · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling · Applied Mathematics · Mathematical Reasoning

Instructions to candidates

- This paper has 45 multiple-choice questions. Each question has four answer choices, of which exactly one is correct.
- Section I (Q1–Q40) carries 2 marks per question. Section II (Q41–Q45) carries 4 marks per question.
- There is no negative marking. Unanswered questions score zero.
- Calculators are not permitted. Rough work may be done on a separate sheet.
- Mark your answers on the answer sheet at the end of the paper.
- All figures are not necessarily drawn to scale unless stated.

For parents and teachers

Print or view this paper in colour. Pictures are part of the questions.

Parts follow the AMO syllabus chapters. Chapter 7 is not part of the Class 7 syllabus, so the chapter numbers shown in the Part headings have gaps.

The answer sheet and the answer key are at the end. Remove only the answer key before the student begins.

Name:

School:

Roll No.:

SECTION I 40 questions × 2 marks each**Part 1 — Number System (Syllabus Chapter 1)**

Q1. On a number line, the point $-\frac{5}{8}$ is marked. Which of these points is to the left of it?

- (A) $-\frac{5}{6}$ (B) $-\frac{1}{2}$ (C) $-\frac{3}{8}$ (D) $-\frac{2}{5}$

Q2. Two points on a number line show $-\frac{5}{6}$ and $\frac{3}{8}$. How far apart are the two points?

- (A) $\frac{29}{24}$ (B) $\frac{4}{7}$ (C) $\frac{11}{24}$ (D) $-\frac{29}{24}$

Q3. At a ferry jetty, a marked pole shows the water level of the river. During a dry spell, the level changes by $-\frac{3}{8}$ m every day. After how many days will the total change in the level be $-2\frac{1}{4}$ m?

- (A) 12 days (B) 3 days (C) 6 days (D) 18 days

Q4. How many numbers of the form $\frac{p}{10}$, where p is an integer, are greater than $-\frac{2}{5}$ and less than $\frac{3}{10}$?

- (A) 5 (B) 6 (C) 7 (D) 8

Part 2 — Operations & Number Properties (Syllabus Chapter 2)

Q5. Which of these numbers is the greatest?

- (A) 2.5×10^6 (B) 9.8×10^5
(C) 1.9×10^6 (D) 6.4×10^5

Q6. The number 3^5 is how many times the number 3^2 ?

- (A) 3 (B) 9 (C) 234 (D) 27

Q7. What is $(3.5 \times 10^4) + (2.5 \times 10^3)$, written in standard form?

- (A) 2.85×10^3 (B) 3.75×10^4
(C) 6×10^4 (D) 6×10^7

Part 3 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

Q8. A shop receives 450 eggs packed in egg trays. On the way, 6% of the eggs crack. How many eggs do NOT crack?

- (A) 444 eggs (B) 27 eggs (C) 423 eggs (D) 405 eggs

Q9. A shopkeeper bought an umbrella stand for ₹1,250. He sold it for ₹1,000. What was his loss percent?

- (A) 2.5% (B) 80% (C) 20% (D) 25%

Q10. In a bookshop sale, every book is sold at 20% off its marked price. Komal buys two books marked ₹250 and ₹350. How much does she pay in all?

- (A) ₹480 (B) ₹550 (C) ₹580 (D) ₹120

Q11. There were two candidates in a school council election. Every student voted for exactly one of them. The winner got 58% of the votes and won by 112 votes. How many students voted?

- (A) 294 (B) 700 (C) 1,400 (D) 406

Q12. Imran's savings are 25% more than Zara's savings. By what percentage are Zara's savings less than Imran's savings?

- (A) 75% (B) 25% (C) 80% (D) 20%

Part 4 — Algebra (Syllabus Chapter 4)

Q13. In an art class, Ojas has n crayons. Rima has 5 fewer than three times as many crayons as Ojas. How many crayons do they have together?

- (A) $4n - 15$ (B) $4n + 5$ (C) $3n - 5$ (D) $4n - 5$

Q14. A wooden plank is $(5x + 3)$ cm long. Two pieces, each $(x - 1)$ cm long, are cut from it to make a shelf. What length of the plank is left?

- (A) $(3x + 5)$ cm (B) $(3x + 1)$ cm
(C) $(7x + 1)$ cm (D) $(4x + 4)$ cm

Q15. Vivek solved the equation $4x - 7 = 2x + 9$ as shown.

Step 1: $4x - 2x - 7 = 9$

Step 2: $2x - 7 = 9$

Step 3: $2x = 9 - 7$

Step 4: $x = 1$

In which step did he make his first mistake?

- (A) Step 1 (B) Step 2 (C) Step 3 (D) Step 4

Q16. Three classes collected 102 kg of old paper for recycling. Class 7B collected 6 kg more than Class 7A. Class 7C collected twice as much as Class 7A. How much paper did Class 7C collect?

- (A) 54 kg (B) 48 kg (C) 30 kg (D) 24 kg

Q17. Vihaan thinks of a number. He multiplies it by 5 and then subtracts 12. Chitra takes the same number, multiplies it by 3 and then adds 10. They both get the same answer. What is the number?

- (A) -1 (B) 43 (C) 11 (D) 22

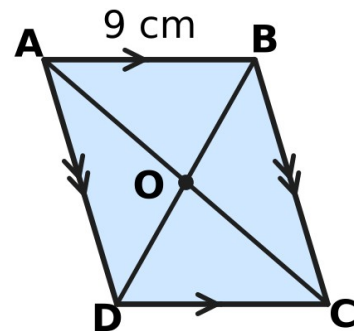
Part 5 — Geometry (Syllabus Chapter 5)

Q18. The sides of a triangle are 7 cm, 12 cm and x cm, where x is a whole number. What is the smallest possible value of x ?

- (A) 18 (B) 19 (C) 6 (D) 5

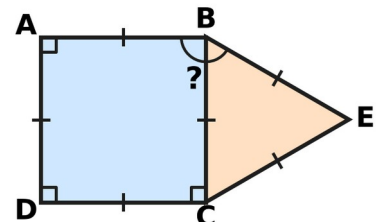
Q19. In the figure, ABCD is a parallelogram. Its diagonals AC and BD meet at O. $AC = 16$ cm, $BD = 12$ cm and $AB = 9$ cm. What is the perimeter of $\triangle AOB$?

- (A) 14 cm (B) 23 cm (C) 29 cm (D) 37 cm



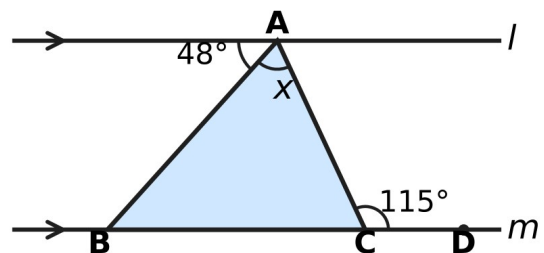
Q20. In the figure, ABCD is a square and BCE is an equilateral triangle drawn outside the square on the side BC. What is the size of $\angle ABE$ (marked "?")?

- (A) 150° (B) 120° (C) 135° (D) 30°



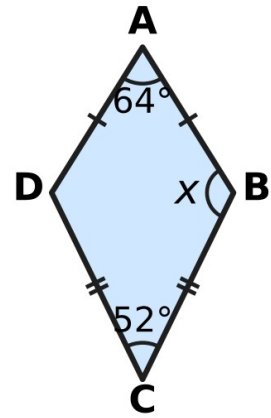
Q21. In the figure, line l is parallel to line m . A is on l , and B, C and D are on m . AB makes an angle of 48° with l , and $\angle ACD = 115^\circ$. What is the value of x ?

- (A) 65° (B) 132° (C) 17° (D) 67°



Q22. In the quadrilateral ABCD shown, $AB = AD$ and $CB = CD$, as marked. $\angle BAD = 64^\circ$ and $\angle BCD = 52^\circ$. What is the value of x , the size of $\angle ABC$?

- (A) 122° (B) 32° (C) 116° (D) 244°



Part 6 — Mensuration (Syllabus Chapter 6)

Q23. A cuboid of modelling clay measures $16 \text{ cm} \times 4 \text{ cm} \times 1 \text{ cm}$. All of it is reshaped into one cube, with no clay left over. What is the length of an edge of the cube?

- (A) 8 cm (B) 7 cm (C) 4 cm (D) 16 cm

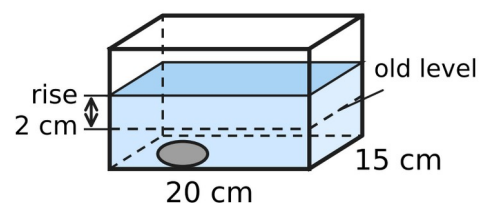
Container	Radius of base (cm)	Height (cm)
Cylinder A	3	8
Cylinder B	5	3
Cylinder C	2	16
Cylinder D	4	5

Q24. The table gives the radius and height of four cylindrical containers. Which container holds the most water when full?

- (A) Cylinder A (B) Cylinder B
(C) Cylinder C (D) Cylinder D

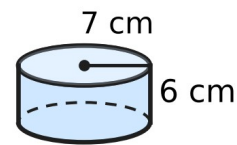
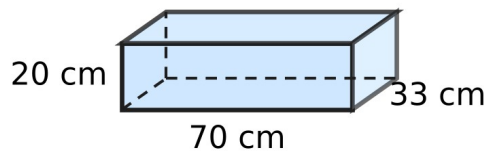
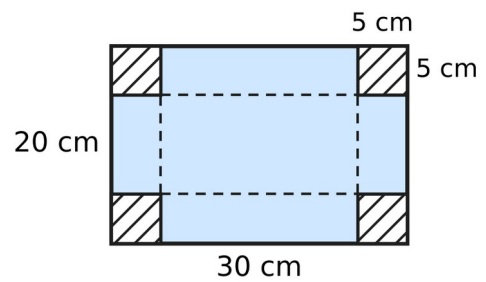
Q25. A cuboid tank has a base 20 cm long and 15 cm wide, and holds some water. A stone is put in the tank. It sinks fully and the water level rises by 2 cm, as shown. No water spills out. What is the volume of the stone?

- (A) 300 cm^3 (B) 600 cm^3 (C) 40 cm^3 (D) 30 cm^3



Q26. A sheet of card is 30 cm long and 20 cm wide. A square of side 5 cm is cut from each corner, as shown. The sides are folded up along the dashed lines to make an open box. What is the volume of the box?

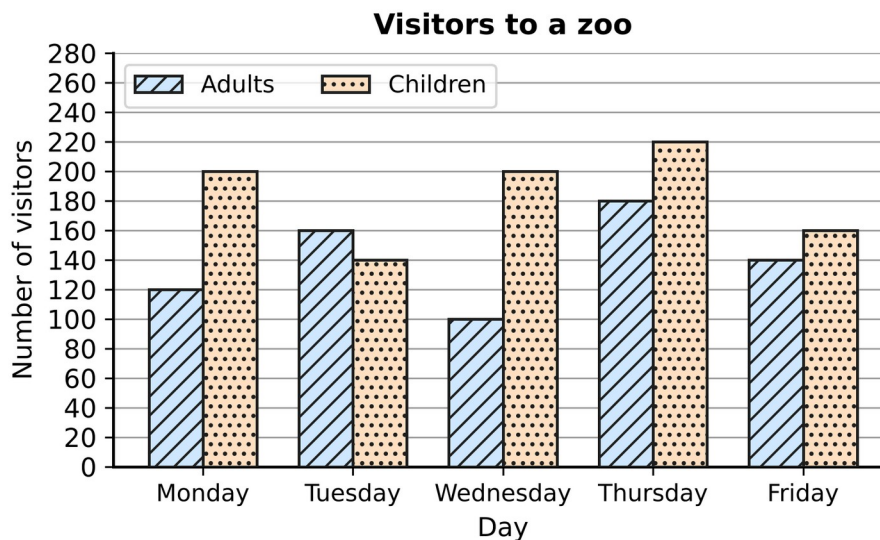
- (A) 1,500 cm³ (B) 1,875 cm³
 (C) 3,000 cm³ (D) 1,000 cm³



Q27. At a milk dairy, the cuboid container shown, 70 cm long, 33 cm wide and 20 cm high, is full of milk. All the milk is poured into cylindrical tubs like the one shown, of radius 7 cm and height 6 cm. How many tubs can be filled completely? Use $\pi = \frac{22}{7}$.

- (A) 200 (B) 300 (C) 175 (D) 50

Part 7 — Data Handling, Statistics & Probability (Syllabus Chapter 8)



Q28. The double bar graph shows the number of adults and children who visited a zoo on five days. On which day was the difference between the number of children and the number of adults the greatest?

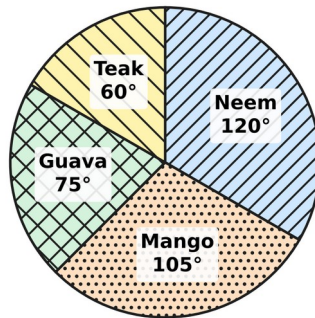
- (A) Monday (B) Wednesday
 (C) Thursday (D) Friday

Number of pets	0	1	2	3	4
Number of students	9	14	10	5	2

Q29. Each of the 40 students of a class wrote down how many pets they have. The table shows the results. What is the mode of the number of pets?

- (A) 2 (B) 4 (C) 1 (D) 14

Saplings sold by a plant nursery in one month



Q30. The pie chart shows the saplings sold by a plant nursery in one month. The angle of each sector is marked. The nursery sold 40 more mango saplings than guava saplings. How many saplings did it sell in all?

- (A) 480 (B) 270 (C) 140 (D) 360

Part 8 — Applied Mathematics (Syllabus Chapter 9)

Q31. Kriti's recipe for a rice dish uses 450 g of rice for 6 people. How much rice does she need to make the same dish for 10 people?

- (A) 75 g (B) 750 g (C) 4,500 g (D) 270 g

Q32. At a school assembly, the students stand in 36 rows with 20 students in each row. The next day the same students stand in rows of 30. How many rows are there now?

- (A) 26 (B) 54 (C) 30 (D) 24

Pack	Mass	Price
Pack P	500 g	₹60
Pack Q	750 g	₹84
Pack R	1 kg	₹110
Pack S	2 kg	₹210

Q33. The table shows four packs of the same washing powder. Which pack has the lowest price for each kilogram?

- (A) Pack P (B) Pack Q (C) Pack R (D) Pack S

Q34. On a mountain hike, 12 hikers need 18 kg of food for 5 days. Every hiker eats the same amount each day. How much food do 8 hikers need for 6 days?

- (A) 32.4 kg (B) 14.4 kg (C) 10 kg (D) 12 kg

Q35. For a school picnic, a teacher buys apples at ₹150 per kg and pears at ₹120 per kg. The masses of apples and pears are in the ratio 4 : 3. She pays ₹2,880 in all. What mass of pears does she buy?

- (A) 12 kg (B) 21 kg (C) 9 kg (D) 3 kg

Part 9 — Mathematical Reasoning & Problem-Solving (Syllabus Chapter 10)

Q36. Look at this pattern.

$$1 \times 8 + 1 = 9$$

$$12 \times 8 + 2 = 98$$

$$123 \times 8 + 3 = 987$$

$$1,234 \times 8 + 4 = 9,876$$

The pattern goes on in the same way. What is $123,456 \times 8 + 6$?

- (A) 987,654 (B) 987,648
(C) 9,876,543 (D) 98,765

Q37. A palindrome is a number that reads the same from left to right and from right to left, such as 363. How many three-digit palindromes are divisible by 5?

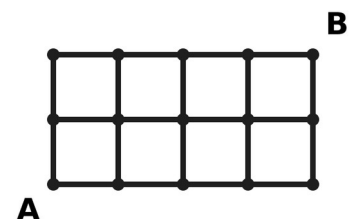
- (A) 10 (B) 9 (C) 20 (D) 90

Q38. Look at this list of numbers: 2, 5, 10, 17, 26, ... The pattern goes on in the same way. Which term of the list is 101?

- (A) 11th term (B) 10th term
(C) 9th term (D) 50th term

Q39. The figure shows the straight paths in a park. Tia walks along the paths from A to B. At every step she goes either to the right or up, never left or down. How many different routes can she take?

- (A) 10 (B) 6 (C) 15 (D) 8



1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Q40. The counting numbers are written in rows of 7, as shown. The grid goes on in the same way. The four numbers inside a 2×2 square are added, for example $2 + 3 + 9 + 10 = 24$. For another 2×2 square, the total is 116. What is the smallest number in that square?

- (A) 33 (B) 29 (C) 26 (D) 25

SECTION II — Star Questions 5 questions $\times$ 4 marks each

Q41. A bag holds only red counters and blue counters. The number of red counters is 5 more than twice the number of blue counters. There are more than 30 and fewer than 40 counters in the bag. All the counters can be shared equally among 4 children, with none left over. How many red counters are in the bag? [4 marks]

- (A) 9 (B) 23 (C) 25 (D) 27

Q42. A wall clock is marked ₹250. The shop may give a discount of a whole-number percentage that is greater than 10% and less than 30%. For how many of these discounts is the sale price a whole number of rupees? [4 marks]

- (A) 19 (B) 3 (C) 9 (D) 10

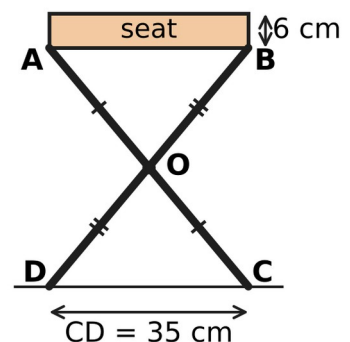
Q43. Garima makes big candles and small candles from the same wax. The mass of 3 big candles is the same as the mass of 5 small candles. A small candle has a mass of 60 g. She packs a box with big and small candles. The box has twice as many small candles as big candles. Together they have a mass of 2,640 g. How many candles are in the box? [4 marks]

- (A) 24 (B) 44 (C) 12 (D) 36

Q44. The figure shows the side view of a folding stool. The legs AC and BD are straight and cross at O. Equal marks show $OA = OC$ and $OB = OD$. The lower ends C and D of the legs are 35 cm apart. The seat is a cylinder 6 cm thick, and AB is a diameter of the seat. What is the volume of the seat?

Use $\pi = \frac{22}{7}$. [4 marks]

- (A) $5,775 \text{ cm}^3$ (B) 660 cm^3
(C) $11,550 \text{ cm}^3$ (D) $23,100 \text{ cm}^3$



Q45. A rational number x is greater than $-\frac{2}{3}$ and less than $-\frac{1}{3}$. On the number line, x and $4x + \frac{5}{2}$ are at the same distance from 0. What is the value of x ? [4 marks]

(A) $-\frac{2}{5}$

(B) $-\frac{5}{8}$

(C) $-\frac{5}{6}$

(D) $-\frac{1}{2}$

Answer Sheet AMO Level 1 • Class 7 • Practice Paper 4**Name:****School:****Roll No.:***Shade or tick one circle for each question.*

- | | | |
|---------------------|---------------------|---------------------|
| 1. (A) (B) (C) (D) | 16. (A) (B) (C) (D) | 31. (A) (B) (C) (D) |
| 2. (A) (B) (C) (D) | 17. (A) (B) (C) (D) | 32. (A) (B) (C) (D) |
| 3. (A) (B) (C) (D) | 18. (A) (B) (C) (D) | 33. (A) (B) (C) (D) |
| 4. (A) (B) (C) (D) | 19. (A) (B) (C) (D) | 34. (A) (B) (C) (D) |
| 5. (A) (B) (C) (D) | 20. (A) (B) (C) (D) | 35. (A) (B) (C) (D) |
| 6. (A) (B) (C) (D) | 21. (A) (B) (C) (D) | 36. (A) (B) (C) (D) |
| 7. (A) (B) (C) (D) | 22. (A) (B) (C) (D) | 37. (A) (B) (C) (D) |
| 8. (A) (B) (C) (D) | 23. (A) (B) (C) (D) | 38. (A) (B) (C) (D) |
| 9. (A) (B) (C) (D) | 24. (A) (B) (C) (D) | 39. (A) (B) (C) (D) |
| 10. (A) (B) (C) (D) | 25. (A) (B) (C) (D) | 40. (A) (B) (C) (D) |
| 11. (A) (B) (C) (D) | 26. (A) (B) (C) (D) | 41. (A) (B) (C) (D) |
| 12. (A) (B) (C) (D) | 27. (A) (B) (C) (D) | 42. (A) (B) (C) (D) |
| 13. (A) (B) (C) (D) | 28. (A) (B) (C) (D) | 43. (A) (B) (C) (D) |
| 14. (A) (B) (C) (D) | 29. (A) (B) (C) (D) | 44. (A) (B) (C) (D) |
| 15. (A) (B) (C) (D) | 30. (A) (B) (C) (D) | 45. (A) (B) (C) (D) |

Q41–Q45 are Section II Star Questions (4 marks each).

For the parent or teacher

Section I correct: _____ / 40 × 2 = _____ marks

Section II correct: _____ / 5 × 4 = _____ marks

Correct: _____ / 45 **Score:** _____ / 100

Answer Key and Solutions AMO Level 1 • Class 7 • Practice Paper 4**Quick answers**

1 A	6 D	11 B	16 B	21 D	26 D	31 B	36 A	41 B
2 A	7 B	12 D	17 C	22 A	27 D	32 D	37 A	42 C
3 C	8 C	13 D	18 C	23 C	28 B	33 D	38 B	43 D
4 B	9 C	14 A	19 B	24 D	29 C	34 B	39 C	44 A
5 A	10 A	15 C	20 A	25 B	30 A	35 C	40 D	45 D

Solutions

Q	Part • level	Answer	Why?
1	Part 1 • Ch1 • Easy	(A) $-\frac{5}{6}$	Points to the left are smaller. $-\frac{3}{8}$, $-\frac{1}{2}$ and $-\frac{2}{5}$ all lie between $-\frac{5}{8}$ and 0. $-\frac{5}{6}$ is less than $-\frac{5}{8}$ (the same 5 parts, but sixths are bigger than eighths), so only $-\frac{5}{6}$ is to the left.
2	Part 1 • Ch1 • Easy	(A) $\frac{29}{24}$	Distance = $\frac{3}{8} - (-\frac{5}{6}) = \frac{3}{8} + \frac{5}{6} = \frac{9}{24} + \frac{20}{24} = \frac{29}{24}$.
3	Part 1 • Ch1 • Medium	(C) 6 days	Number of days = $(-\frac{9}{4}) \div (-\frac{3}{8}) = (-\frac{9}{4}) \times (-\frac{8}{3}) = \frac{72}{12} = 6$.
4	Part 1 • Ch1 • Medium	(B) 6	$-\frac{2}{5} = -\frac{4}{10}$. The numbers $\frac{p}{10}$ with $-4 < p < 3$ have $p = -3, -2, -1, 0, 1, 2$. That is 6 numbers.
5	Part 2 • Ch2 • Easy	(A) 2.5×10^6	Numbers with 10^6 are greater than numbers with 10^5 . Of 2.5×10^6 and 1.9×10^6 , the greater is 2.5×10^6 (2,500,000).
6	Part 2 • Ch2 • Easy	(D) 27	We need $3^5 \div 3^2 = 3^{5-2} = 3^3 = 27$. Check: $243 \div 9 = 27$.
7	Part 2 • Ch2 • Medium	(B) 3.75×10^4	Write 2.5×10^3 as 0.25×10^4 . So the sum is $(3.5 + 0.25) \times 10^4 = 3.75 \times 10^4$ (that is, $35,000 + 2,500 = 37,500$).
8	Part 3 • Ch3 • Easy	(C) 423 eggs	The cracked eggs are 6% of 450 = $450 \times \frac{6}{100} = 27$. So $450 - 27 = 423$ eggs do not crack.
9	Part 3 • Ch3 • Easy	(C) 20%	Loss = $1,250 - 1,000 = ₹250$. Loss percent = $\frac{250}{1,250} \times 100 = 20\%$.
10	Part 3 • Ch3 • Easy	(A) ₹480	Total marked price = $250 + 350 = ₹600$. Discount = 20% of 600 = ₹120. She pays $600 - 120 = ₹480$.
11	Part 3 • Ch3 • Medium	(B) 700	The other candidate got 42%. The winning margin is $58\% - 42\% = 16\%$ of the votes = 112. So 1% is 7 votes and the total is 700.
12	Part 3 • Ch3 • Medium	(D) 20%	If Zara has ₹100, Imran has ₹125. Zara has ₹25 less, and $\frac{25}{125} \times 100 = 20\%$.
13	Part 4 • Ch4 • Easy	(D) $4n - 5$	Rima has $3n - 5$ crayons. Together they have $n + (3n - 5) = 4n - 5$.
14	Part 4 • Ch4 • Easy	(A) $(3x + 5)$ cm	Left = $(5x + 3) - 2(x - 1) = 5x + 3 - 2x + 2 = (3x + 5)$ cm.
15	Part 4 • Ch4 • Easy	(C) Step 3	From $2x - 7 = 9$, adding 7 to both sides gives $2x = 16$, not $9 - 7$. So Step 3 is wrong; the correct answer is $x = 8$.
16	Part 4 • Ch4 • Medium	(B) 48 kg	Let Class 7A collect x kg. Then $x + (x + 6) + 2x = 102$, so $4x = 96$ and $x = 24$. Class 7C collected $2 \times 24 = 48$ kg.
17	Part 4 • Ch4 • Medium	(C) 11	Let the number be n . Then $5n - 12 = 3n + 10$, so $2n = 22$ and $n = 11$. Check: $55 - 12 = 43$ and $33 + 10 = 43$.
18	Part 5 • Ch5 • Easy	(C) 6	The sum $7 + x$ must be greater than 12, so x is greater than 5. The smallest whole number greater than 5 is 6 (and 6, 7, 12 do make a triangle).

Q	Part · level	Answer	Why?
19	Part 5 · Ch5 · Easy	(B) 23 cm	The diagonals of a parallelogram bisect each other, so $AO = 8$ cm and $BO = 6$ cm. Perimeter = $8 + 6 + 9 = 23$ cm.
20	Part 5 · Ch5 · Easy	(A) 150°	$\angle ABC = 90^\circ$ (angle of a square) and $\angle CBE = 60^\circ$ (angle of an equilateral triangle). $\angle ABE = 90^\circ + 60^\circ = 150^\circ$.
21	Part 5 · Ch5 · Medium	(D) 67°	$\angle ABC = 48^\circ$ (alternate angles, $l \parallel m$). The exterior angle $\angle ACD$ equals $\angle ABC + x$, so $x = 115^\circ - 48^\circ = 67^\circ$.
22	Part 5 · Ch5 · Medium	(A) 122°	$\triangle ABC \cong \triangle ADC$ by SSS ($AB = AD$, $CB = CD$, AC common), so $\angle ABC = \angle ADC$. These two add up to $360^\circ - 64^\circ - 52^\circ = 244^\circ$, so $x = 122^\circ$.
23	Part 6 · Ch6 · Easy	(C) 4 cm	Volume = $16 \times 4 \times 1 = 64$ cm ³ . The cube has the same volume, and $4 \times 4 \times 4 = 64$, so its edge is 4 cm.
24	Part 6 · Ch6 · Easy	(D) Cylinder D	Volume = $\pi r^2 h$, so compare $r \times r \times h$: A 72, B 75, C 64, D 80 (each times π). Cylinder D holds the most.
25	Part 6 · Ch6 · Easy	(B) 600 cm ³	The stone pushes up a layer of water 20 cm $\times$ 15 cm $\times$ 2 cm. Its volume = $20 \times 15 \times 2 = 600$ cm ³ .
26	Part 6 · Ch6 · Medium	(D) 1,000 cm ³	The base of the box is $(30 - 10)$ cm by $(20 - 10)$ cm = 20 cm by 10 cm, and the height is 5 cm. Volume = $20 \times 10 \times 5 = 1,000$ cm ³ .
27	Part 6 · Ch6 · Medium	(D) 50	Milk = $70 \times 33 \times 20 = 46,200$ cm ³ . One tub holds $22/7 \times 7 \times 7 \times 6 = 924$ cm ³ . Number of tubs = $46,200 \div 924 = 50$.
28	Part 7 · Ch8 · Easy	(B) Wednesday	Differences: Monday 80, Tuesday 20, Wednesday 100, Thursday 40, Friday 20. The greatest is on Wednesday.
29	Part 7 · Ch8 · Easy	(C) 1	The mode is the value that occurs most often. There are 14 students with 1 pet, more than for any other value, so the mode is 1.
30	Part 7 · Ch8 · Medium	(A) 480	The difference between the sectors is $105^\circ - 75^\circ = 30^\circ$, which stands for 40 saplings. So 360° stands for $40 \times 12 = 480$ saplings.
31	Part 8 · Ch9 · Easy	(B) 750 g	Rice for 1 person = $450 \div 6 = 75$ g. For 10 people: $75 \times 10 = 750$ g.
32	Part 8 · Ch9 · Easy	(D) 24	Number of students = $36 \times 20 = 720$. In rows of 30: $720 \div 30 = 24$ rows.
33	Part 8 · Ch9 · Easy	(D) Pack S	Price per kg: P ₹120, Q ₹112, R ₹110, S ₹105. Pack S is the cheapest for each kilogram.
34	Part 8 · Ch9 · Medium	(B) 14.4 kg	Food for 1 hiker for 1 day = $18 \div (12 \times 5) = 0.3$ kg. For 8 hikers for 6 days: $0.3 \times 8 \times 6 = 14.4$ kg.
35	Part 8 · Ch9 · Medium	(C) 9 kg	Then 4 kg of apples and 3 kg of pears cost $4 \times 150 + 3 \times 120 = ₹960$. Since $₹2,880 \div ₹960 = 3$, she buys $3 \times 3 = 9$ kg of pears (and 12 kg of apples).
36	Part 9 · Ch10 · Easy	(A) 987,654	In each line the answer counts down from 9, with as many digits as the first number has. The number 123,456 has six digits, so the answer is 987,654.
37	Part 9 · Ch10 · Easy	(A) 10	A number divisible by 5 ends in 0 or 5. A palindrome ending in 0 would start with 0, so it must be 5_5. The middle digit can be 0 to 9: 10 palindromes.
38	Part 9 · Ch10 · Easy	(B) 10th term	Each term is one more than a square: $1^2 + 1$, $2^2 + 1$, $3^2 + 1$, ... Since $101 = 10^2 + 1$, it is the 10th term.
39	Part 9 · Ch10 · Medium	(C) 15	Write at each corner the number of ways to reach it: it is the sum of the numbers just to the left and just below. The bottom row and left column are all 1; the top row becomes 1, 3, 6, 10, 15. So there are 15 routes.

Q	Part • level	Answer	Why?
40	Part 9 • Ch10 • Medium	(D) 25	If the smallest number is n , the square holds n , $n + 1$, $n + 7$ and $n + 8$, with total $4n + 16$. So $4n + 16 = 116$ and $n = 25$ (square 25, 26, 32, 33).
41	Section II • Ch4 + Ch10 • Hard	(B) 23	With b blue counters there are $2b + 5$ red ones, so $3b + 5$ in all. Between 30 and 40 this gives $b = 9, 10$ or 11 (totals 32, 35, 38). Only 32 is divisible by 4, so $b = 9$ and there are 23 red counters.
42	Section II • Ch3 + Ch10 • Hard	(C) 9	A discount of $d\%$ of ₹250 is $250 \times d/100 = 5d/2$ rupees. This is a whole number only when d is even. The even values greater than 10 and less than 30 are 12, 14, ..., 28: 9 discounts.
43	Section II • Ch4 + Ch9 • Hard	(D) 36	A big candle has mass $5 \times 60 \div 3 = 100$ g. With x big and $2x$ small candles: $100x + 120x = 2,640$, so $220x = 2,640$ and $x = 12$. The box has $12 + 24 = 36$ candles.
44	Section II • Ch5 + Ch6 • Hard	(A) $5,775 \text{ cm}^3$	$\angle AOB = \angle COD$ (vertically opposite), $OA = OC$ and $OB = OD$, so $\triangle AOB \cong \triangle COD$ (SAS) and $AB = CD = 35$ cm. The radius is 17.5 cm. Volume = $22/7 \times 17.5 \times 17.5 \times 6 = 5,775 \text{ cm}^3$.
45	Section II • Ch1 + Ch4 • Hard	(D) $-\frac{1}{2}$	Numbers at the same distance from 0 are equal or opposite. If $4x + 5/2 = x$, then $x = -5/6$, which is less than $-2/3$, so it does not fit. If $4x + 5/2 = -x$, then $5x = -5/2$, so $x = -1/2$, which is between $-2/3$ and $-1/3$.